PS-1 run report: algebra spec, bit-degree and root-test formulations

PDP oracle speedup program, `research/pdp_oracle_speedup_20261011/ps1_algebra/` — run of 2026-10-11, 22:17 UTC

1. What ran

`ps1_algebra/run_ps1.sh` with `examples/pdp_speedup` (subcommands `degrees`, `torsion`, `agree`, `bs-check`, `sample`, `corpus-check`, `gates`), Conductor task T-4, on the PS-0 host (Intel Xeon 2.80 GHz KVM guest, AVX-512, no GPU). No timing is taken; every figure is a count or an exact comparison. Raw records: `ps1_algebra/raw/*.jsonl`.

2. Observations

Bit-degree (measured, $l = 3..8$, six targets each). Joint algebraic degree of the output bits of $c_0..c_4$ in the bits of (x_1, x_2): 2, 4, 2, 4, 2 at every l; in the bits of x_2 alone with x_1 fixed: 1, 2, 1, 2, 1.

Exact agreement (summation-polynomial roots against the group-law reference).

check	cases	items	disagreements
exhaustive, fields of degree 5–9, all $l \leq n$, all rational targets	35	4168 targets, 15,628,190 roots	0 at non-degenerate targets; 1300 explained extra roots at degenerate targets; 0 unexplained
bit-sliced 64-lane pipeline against exact reference, same range	35	2834 targets, 1,259,432 pairs	0 (10,814 zero-denominator pairs routed to the exact path, not compared)
sampled, $(n, l) \in \{(11,3),(11,4),(11,5),(13,4),(13,5),(13,6)\}$	6	6×10^6 random pairs	0
PS-0 corpus root sets, $l = 8..12$	5	400 targets	0
translation by the 2-torsion point is $x \mapsto \frac{1}{x}$, degrees 5–12 exhaustive, 24–36 at 200,000 samples	13	about 1.0 million points	0

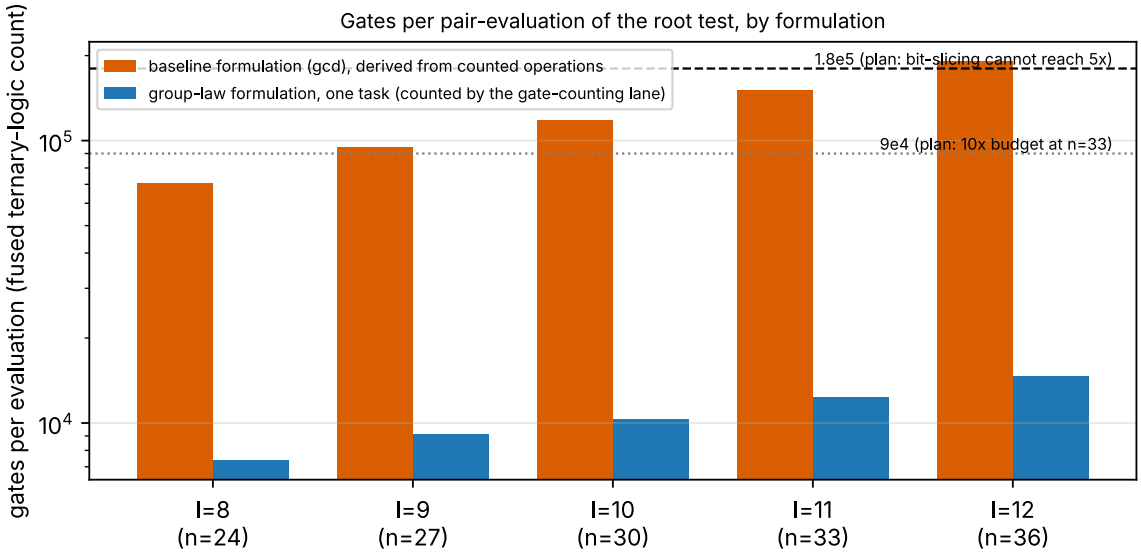


Figure 1: Gates per evaluation of the root test, by formulation and rung (source `ps1_algebra/raw/gates.jsonl`; fused ternary-logic count). The baseline formulation is derived from PS-0's counted operations; the group-law task is counted by running the same code on a gate-counting lane. Log scale.

l	n	group-law task, fused	unfused	baseline (derived)	ratio
8	24	7,410	12,766	70,695	9.5
9	27	9,137	15,961	94,729	10.4
10	30	10,354	18,869	118,099	11.4
11	33	12,349	22,653	151,371	12.3
12	36	14,608	27,011	190,275	13.0

3. Missing or not done

No `identity.certificate/v1` record exists: the schema is fixed to F_p with $p = 2^{61} - 1$ and cannot express a characteristic-2 identity. The baseline formulation's count is derived from counted operations, not a built program. The trace-based and resolvent-cubic candidates were not attempted. The sampled comparison at $n = 11, 13$ is against the generic summation-polynomial enumerator, not against `decompose`. Wall time was not measured; the group-law test is a different algorithm, valid for targets outside $x_R \in V \cup V^{-1} \cup \{0\}$.